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AI-driven Optimization of project portfolios in corporate ecosystems with synergies and strategic factors

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ABSTRACT

This paper studies the optimization of project portfolios in corporate ecosystems by considering both strategic factors and return synergies between projects. We propose a hybrid method that combines machine learning with mathematical programming to address this enhanced form of project portfolio optimization. Unlike traditional approaches, which evaluate projects mainly based on individual risks and returns, our framework considers strategic priorities and the extra value created when projects reinforce each other. Machine learning models predict synergies, while exact optimization ensures consistent portfolio selection under resource and strategic constraints. A numerical proof-of-concept illustrates the methodology. Computational experiments show that portfolios designed with synergy and strategy in mind might achieve a significantly higher performance than portfolios that do not account for project synergies. The paper also examines computational efficiency and scalability, highlighting the approach's potential for practical application in complex and dynamic corporate ecosystems.

1. Introduction

A corporate ecosystem is a dynamic network of companies, units, and partners that co-create value through shared resources, complementary capabilities, and coordinated strategies. These ecosystems allow firms to move beyond traditional organizational boundaries, pooling expertise, accelerating innovation, and adapting more effectively to market changes (Rodriguez-Garcia et al., 2023). With growing interconnectedness and collaboration, they have become increasingly important for sustaining competitive advantage (Jacobides et al., 2018). By facilitating resource and capability sharing across sectors, ecosystems enhance agility, promote innovation, and improve market positioning (Gueler & Schneider, 2021). However, their success depends on effective project portfolio management (Mayer et al., 2017).

Managing ecosystems rather than individual organizations introduces complex challenges, especially when projects span diverse sectors and strategic functions. Traditional portfolio optimization models focus mainly on individual project risk and return and often fail to capture synergies. Here, synergy refers to the additional value created when two or more projects are pursued together due to complementary effects such as resource sharing, capability alignment, or market reinforcement, which enhance combined performance beyond the sum of individual impacts (Saiz et al., 2024). For example, a digital onboarding platform and an AI-driven fraud detection system may each improve an insurer's efficiency, but together they can reduce drop-offs and increase trust, boosting customer acquisition and retention simultaneously. Capturing such synergies is critical, as they can generate economies of scale, expand market reach, and improve adaptability to technological and regulatory changes (Martín-Peña et al., 2024). Existing approaches often fall short due to their static design and limited adaptability to changing conditions (Adner, 2017).

As ecosystems grow in size and complexity, project interactions become more dynamic and nonlinear, further challenging traditional optimization methods (Iansiti, 2004). Ecosystem-based optimization must account explicitly for synergies to maximize collective value and strengthen strategic alignment (Goold & Campbell, 1998). This creates new challenges for CEOs, who must act as coordinators of complex networks, aligning synergy potential with corporate strategic goals and ensuring that selected projects reinforce core capabilities and long-term objectives (Iansiti, 2004). Ecosystem investments form networks of

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interdependencies, as described in systemic risk and network theory models (Acemoglu et al., 2015), where synergies and contagion effects propagate across assets. Traditional optimization approaches often struggle to capture these interdependencies, leading to missed opportunities and inefficient portfolios (Adner, 2017). Moreover, the relevance of synergy effects changes over time in response to technological advances, regulatory shifts, and market evolution (Han et al., 2022).

This paper presents a methodology that combines mathematical optimization with artificial intelligence to optimize the risk-return trade-off in project portfolios while accounting for synergies as well as strategic factors, including desirability, sustainability, and feasibility (DSF). Our approach integrates predictive machine learning and mathematical optimization within a decision-support system. Specifically, we use the an ensemble boosting model (Kunapuli, 2023) to estimate synergy-adjusted project returns, which are then input into a constrained optimization model and solved with an exact solver. Challenges in advanced optimization have been discussed by Koijen et al. (2016), who propose methods to distinguish genuine relationships from spurious correlations. Brandt et al. (2009) present a framework for incorporating firm-level characteristics into portfolio optimization. Their work supports the integration of synergies and strategic factors, reinforcing the relevance of the criteria adopted in our study. The proposed methodology is illustrated through a numerical proof-of-concept, and computational experiments assess its efficiency and scalability. While classical approaches evaluate projects in isolation and rely on static assumptions, our framework captures the dynamic and nonlinear effects of synergies and adapts to changing ecosystem conditions. Combining the predictive power of machine learning with the precision of exact optimization allows us to provide decision-makers with reliable insights into how different project configurations may create additional value and support long-term DSF goals.

The remainder of the paper is structured as follows. Section 2 provides a literature review on project portfolio optimization and its relevance for corporate ecosystems. Section 3 describes the mathematical model, while Section 4 introduces our hybrid solution approach. Section 5 describes a series of computational experiments that contribute to visually illustrating the concepts behind our approach. Section 6 provides a descriptive analysis of the experimental results, while Section 7 offers a more advanced statistical analysis. Section 8 discusses, using numerical experiments, the scalability and robustness of the proposed methodology. Section 9 offers managerial insights. Finally, Section 10 highlights the main conclusions of our study, explains its limitations, and proposes directions for future research.

2. Literature review

Project portfolio optimization in corporate ecosystems has gained increasing attention due to the growing complexity, interdependencies, and dynamic interactions among projects (Torrente & Uberti, 2023). Traditional models focus primarily on individual project risk and return, often assuming independence and static conditions (Liu & Chen, 2018). While these approaches provide a baseline for decision-making, they overlook the additional value created when projects interact synergistically or align strategically across the ecosystem. Synergies can manifest as economies of scale, shared innovation, or market expansion, enhancing portfolio performance beyond the sum of individual project outcomes (Milhomem & Dantas, 2020). Capturing such effects is critical for ecosystem leaders, yet most current optimization frameworks remain static and poorly suited to dynamic interdependencies (Ghanbari et al., 2024).

Several theoretical perspectives provide insight into the emergence and management of synergies. Real options theory frames projects as flexible opportunities that managers may exercise or defer, highlighting the potential value of timely, adaptive decisions (Ioulianou et al., 2021; Kaufmann et al., 2021). Network effects emphasize the amplified value of interconnected projects, particularly in digital and platform-based ecosystems (Estrin et al., 2022). The dynamic capabilities framework focuses on firms' ability to sense opportunities, reconfigure resources, and coordinate actions to realize synergies (Barbosa & Carvalho, 2024; Bechtel et al., 2023). Additionally, project interactions may produce feedback effects, where the outcome of one project influences others within the portfolio, further enhancing or constraining value creation (De Long et al., 1990). These perspectives collectively support the idea that portfolio value in ecosystems is shaped not only by project selection but also by interaction, coordination, and adaptive management.

Practical decision-making in portfolio management often relies on expert judgment, particularly when assessing strategic priorities or estimating project synergies. While indispensable in many contexts, expert-based processes may introduce biases related to evaluators' experience, perspectives, or environmental conditions (Benchimol et al., 2022). Parameter uncertainty is another intrinsic feature of portfolio optimization. Frameworks have been developed to incorporate such uncertainty, improving the robustness of selection decisions under imperfect or incomplete information (Kan & Zhou, 2007). Considering both expert judgment and uncertainty is essential for reliable assessment of potential synergies and strategic fit.

The growing role of AI has opened new avenues for managing portfolios under uncertainty. Studies in financial and industrial contexts show that machine learning methods, including deep reinforcement learning and ensemble models, improve predictive accuracy, enhance risk-return trade-offs, and support adaptive portfolio adjustment under changing conditions (Alfzari et al., 2025; Faheem et al., 2022; Olanrewaju et al., 2025; Roger, 2024). In Operations Research, hybrid frameworks that integrate machine learning predictions with exact optimization have been proposed for applications such as resource allocation, scheduling, and portfolio selection (Nguyen & Bui, 2021; Viana & Matta, 2022; Zhou et al., 2020). These 'predict-then-optimize' approaches use predictive models to estimate key parameters, such as project returns, risks, or interaction effects, which are then incorporated into prescriptive optimization models (Bertsimas & Kallus, 2020; Elmachtoub & Grigas, 2022). Tree-based ensemble methods, particularly gradient boosting, have shown strong performance for structured data due to their ability to capture non-linear relationships and heterogeneous effects (Chen & Guestrin, 2016).

Beyond AI, various advanced optimization techniques have been applied to multi-objective portfolio selection in ecosystems. Genetic algorithms and evolutionary methods have been used to integrate strategic alignment, risk management, and uncertainty handling (Balderas et al., 2019; Ghanbari et al., 2024; Rezahoseini et al., 2020). These approaches improve robustness across multiple criteria but often remain computationally intensive, particularly when modeling complex project interdependencies (Abbasi et al., 2020). Other studies emphasize the importance of modeling project types, interactions, and real-time adaptability to prevent misalignment and suboptimal selection (Nielsen et al., 2024; Panadero et al., 2020).

Despite these advances, significant limitations remain in current portfolio models. Many frameworks do not fully capture evolving synergies, feedback effects, or dynamic interdependencies, and few integrate real-time, data-driven insights into optimization processes (He & Zhang, 2023; Peng & Linetsky, 2024). Even hybrid models that combine exact optimization with machine learning, while promising, are still limited in ecosystem applications (Abbasi et al., 2020; Ghanbari et al., 2024; Saborido et al., 2016; Wang & Aste, 2023). Practical challenges, including cultural integration, incentive alignment, and communication across diverse actors, further complicate the implementation of theoretically sound approaches. Addressing these gaps is essential to create project portfolios that are not only optimized for individual outcomes but also capable of adapting to dynamic conditions, incorporating uncertainty, and realizing the full potential of synergies and feedback effects across complex corporate ecosystems (Gomes et al., 2022; Thornton & Zhao, 2025).

3. Mathematical model

The CEO's problem can be formalized as a rich portfolio optimization problem that accounts for synergies among projects. In this setting, CEOs seek to achieve objectives, such as minimizing risk, while ensuring strategic alignment across their project portfolio. The model considers a set of n projects, $P = \{1, 2, \ldots, n\}$, where the decision variable x_i represents the proportion of the available budget allocated to project i. Unlike traditional approaches, this model introduces synergy effects, which reflect how project interactions enhance overall portfolio value. Synergies are incorporated through an adjustment factor e_i for each project, which modifies its expected return r_i. This synergy-adjusted return is computed as:

$$r_i^{\text{adj}} = r_i \cdot e_i$$

The 'true' synergy score e_i is given by an unknown (black-box) function that emulates reality and depends on project-level indicators, such as desirability (d_i), sustainability (s_i), and technical feasibility (t_i). These indicators capture systemic contributions of each project without modeling explicit pairwise interactions, thus simplifying the optimization problem. From historical data, these synergy scores can also be estimated using a (white-box) predictive model. The optimization objective is to minimize total portfolio risk, represented by the weighted covariance among projects:

$$\text{Minimize:} \quad \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \sigma_{ij}$$

subject to the following constraints:

$$\sum_{i=1}^{n} x_i r_i e_i \geq R \quad \text{(return constraint)} \tag{1}$$

$$\sum_{i=1}^{n} x_i \leq 1 \quad \text{(budget constraint)} \tag{2}$$

$$x_i \geq 0 \quad \forall i \in P \quad \text{(non-negativity constraint)} \tag{3}$$

$$\sum_{i=1}^{n} d_i x_i \geq D \quad \text{(desirability constraint)} \tag{4}$$

$$\sum_{i=1}^{n} s_i x_i \geq S \quad \text{(sustainability constraint)} \tag{5}$$

$$\sum_{i=1}^{n} t_i x_i \geq T \quad \text{(technical feasibility constraint)} \tag{6}$$

$$n_{\min} \leq \sum_{i=1}^{n} z_i \leq n_{\max} \quad \text{(portfolio size constraint)} \tag{7}$$

$$z_i \in \{0, 1\} \quad \forall i \in P \quad \text{(binary decision variables for project selection)} \tag{8}$$

$$l_i z_i \leq x_i \leq u_i z_i \quad \forall i \in P \quad \text{(minimum/maximum investment constraint)} \tag{9}$$

These constraints reflect a broader set of managerial priorities. The desirability, sustainability, and technical feasibility scores ensure that projects are aligned with customer needs, environmental and governance goals, and technical viability (Rodriguez-Garcia et al., 2024). Investment bounds prevent resource over-concentration or excessive fragmentation. The synergy adjustment embedded in the return constraint links financial performance with ecosystem-wide strategic benefits. The optimization outcomes are presented as Pareto curves, showing trade-offs between risk and return. By comparing solutions with and without synergy adjustments, the contribution of synergies to improved strategic alignment and performance becomes evident. This helps decision-makers assess the added value of considering interdependencies in their portfolio selection process.

The model relies on the following assumptions for tractability and interpretability. First, all input metrics (including return, risk, and strategic scores) are known and fixed at the time of optimization. Second, 'true' synergy effects can be efficiently estimated using a predictive model.

4. Solution methodology

We reviewed a set of alternative methodologies commonly found in the literature and practice. Table 1 provides a theoretical comparison of these approaches. These include heuristic baselines (such as random or value-based selection), rule-based filters informed by strategic planning principles, and more advanced AI-based methods using machine learning models for synergy prediction (e.g., XGBoost, random forest, LightGBM). Although these alternatives are not empirically tested in this study, their inclusion helps highlight key design dimensions in project portfolio optimization in corporate ecosystems, where synergies and strategic factors need to be considered. These dimensions include scalability, interpretability, and robustness to data variability. Understanding the trade-offs among these methods supports informed methodological choices for future applications and comparative evaluations.

In this paper, we adopt the XGBoost (extreme gradient boosting) regressor (Deka & Weiner, 2024) for synergy prediction in combination with the Gurobi exact solver (https://www.gurobi.com/) for project portfolio optimization. For the predictive component, similar results could be obtained using LightGBM or random forest regressors, as these are also established tree-based ensemble methods. We selected XGBoost because it has consistently shown strong performance in terms of accuracy and computational efficiency (Choudhury et al., 2024; Yu et al.,

Table 1
Theoretical comparison of potential methodologies.

Methodology	Key characteristics	Type of output	Strengths and limitations	Key references
Random selection	No filtering or prioritization; projects selected randomly	Portfolio selection (binary)	Useful baseline; lacks strategic rationale or synergy consideration	Koza and et al (1992)
Greedy heuristic (value-based)	Projects ranked by standalone value; selected until budget is exhausted	Ranked project list or binary selection	Simple and fast; ignores interdependencies and synergy effects	Ghasemzadeh and Archer (2000), Martino (1995)
Rule-based strategic fit filter	Filters projects based on thresholds (e.g., desirability, feasibility, sustainability)	Reduced project shortlist	Aligns with strategic goals; does not model interactions or optimize globally	Kaplan et al. (1996), Saaty and Vargas (1980), Levine (2005), Cooper et al. (2001)
XGBoost + exact solver	Synergies predicted via XGBoost; exact solver selects optimal portfolio under constraints	Near-optimal portfolio selection	Interpretable and data-driven; sensitive to model assumptions; no uncertainty modeling	Chen and Guestrin (2016), Bertsimas and Tsitsiklis (1997)
Random forest + exact solver	Random forest predicts synergies; exact solver used for optimization	Near-optimal portfolio selection	May improve robustness to noise; reduced interpretability; parameter tuning needed	Breiman (2001), Hastie et al. (2009)
LightGBM + exact solver	LightGBM estimates synergies; exact solver optimizes portfolio	Near-optimal portfolio selection	High efficiency on large datasets; may underperform on small samples	Ke et al. (2017)

2025), making it a practical choice for our setting. For the optimization component, other exact solvers such as CPLEX (https://www.ibm.com/products/ilog-cplex-optimization-studio) would likely produce equivalent results in our proof-of-concept tests, since both solvers provide highly optimized implementations of the same core algorithms.

Hence, in order to solve a rich project portfolio optimization problem with synergy effects and strategic factors, we propose an algorithm that combines an exact solver with machine learning. The algorithm relies on two main components: (i) predictive modeling using XGBoost to estimate synergy-adjusted returns; and (ii) a Pyomo-Gurobi optimization framework to solve the resulting deterministic optimization problem under complex constraints. Pyomo is a Python-based modeling language for formulating mathematical optimization problems (https://www.pyomo.org/). As already explained, Gurobi is an exact solver for mathematical programming known for speed and efficiency in solving mixed-integer linear programs (Lin et al., 2025). For predictive modeling, we use XGBoost, an algorithm effective in regression and classification tasks (Fatima et al., 2023). To avoid overfitting, we apply early stopping during training by monitoring performance on a validation set and halting training once performance stops improving (Deka & Weiner, 2024). By integrating machine learning and mathematical programming, our methodology makes use of the strengths of both approaches to optimize the portfolio while accounting for synergy effects. Algorithm 1 outlines the basic steps of the proposed methodology.

Algorithm 1 Algorithm for project portfolio with synergy effects.

Input: Project features (desirability, sustainability, feasibility), budget, constraints
Generate synthetic training dataset simulating project interactions
Compute DSF values for baseline feature expectations
Step 1: Baseline Optimization
Formulate baseline optimization problem (Pyomo):
    Decision variables: investment levels, project selections
    Objective: minimize portfolio risk
    Constraints: budget, expected returns, DSF thresholds, cardinality, investment bounds
Solve optimization problem (Gurobi)
Step 2: Synergy Modeling
Train regression model (XGBoost model with early stopping) on synergies dataset
Predict synergy score e_i for each project i
Step 3: Synergy-Aware Optimization
Adjust project returns using synergy scores
Update optimization problem parameters with synergy-adjusted returns
Solve updated optimization problem (Gurobi):
Output: Optimal portfolio considering synergy effects and DSF constraints

For each DSF feature, we compute average values to define baseline expectations across the project set. The Pyomo-Gurobi framework constructs a base Pareto frontier using expected return, risk, and DSF values. Decision variables represent investment levels and project selections. The model minimizes portfolio risk under constraints that include: total investment budget, minimum DSF thresholds, project count limits (cardinality), and bounds on individual investments. Solver settings include: 300-second time limit per run, 0.01 % optimality gap, 400 scenario evaluations, selection of 3 to 7 projects (user-defined bounds for the size of the project portfolio), and strategic constraints on DSF. At this stage, synergy effects are not yet modeled, while inputs are treated as deterministic.

In the second step, we incorporate synergy effects, which can be both positive or negative. An XGBoost model is trained on a dataset of historical projects with synergies. The model is configured with: learning rate = 0.1, max depth = 5, early stopping rounds = 20, evaluation metric = RMSE, 5-fold cross-validation, and squared error as the objective function. Positive synergies are assumed when projects share high DSF values, while negative synergies reflect divergence in key features. Once trained, the model estimates a synergy score e_i for each project $i \in P$. These scores are used to adjust the expected returns and are passed as parameters to the optimization phase. The optimization structure remains unchanged, but now includes synergy-adjusted returns. Gurobi solves the updated problem using the same configuration as before.

5. Proof-of-concept experiments

The proposed algorithm is implemented in Python 3.11, using packages such as Pyomo to define the objective function, constraints, and decision variables. These initial tests were run on a standard personal computer equipped with an Intel Core *i*7 CPU at 2.5 GHz and 12 GB of RAM, running Windows 10. The hybrid framework combines Pyomo 6.5, Gurobi 10.0, and XGBoost 1.7.4.

Data was collected from companies listed on the Nasdaq in the United States via Datastream, covering the period from March 2022 to March 2024. The Nasdaq index was selected for three main reasons: (i) it offers high-frequency, time-series data across its listed constituents, which is essential for reliably estimating dynamic linkages; (ii) it covers a broad yet balanced mix of sectors, avoiding extreme concentration or sparsity that can complicate network calibration; and (iii) its data are publicly accessible and consistently reported, supporting reproducibility of our proof-of-concept. Each company is treated as a project within the potential corporate ecosystem. The initial database includes 1,782 records from 38 countries, spanning 12 sectors and 138 industries. Although the dataset is diverse across sector, industry, and company size (capitalization), 87 % of the data is concentrated in the United States. A thorough data cleaning process was applied to remove anomalies and incomplete records, reducing the dataset by 9.9 % to 1,605 high-quality records. To ensure alignment with the home ecosystem context, we decided to exclude sectors and companies that we considered unsuitable for our study, such as basic materials, mining, paper, military, biological products, and data processing technology. This further reduced the dataset by 43.2 %, resulting in 911 records.

Return is calculated as the average over 34 periods, each spanning 3 weeks, covering a total of 2 years. This smooths out short-term fluctuations and provides a robust measure of company returns. Risk is measured by standard deviation, capturing variability in project returns. Due to the absence of publicly available objective data for the DSF dimensions, these were constructed using structured heuristics based on sector characteristics, environmental, social, and governance (ESG) insights, and market proxies to simulate realistic strategic constraints. The desirability score, scaled from 0 to 1, reflects customer value perception and interest. Sectors such as advertising and public utilities (water, energy, light, gas) received higher scores associated with high perceived interest, while vehicle manufacturing and tobacco were rated lower. Feasibility, also on a 0 to 1 scale, assesses the ease of forming alliances across sectors, considering factors like value alignment, corporate culture, and technological compatibility. Companies were categorized by market capitalization, with very small (< \$150 million) or extremely large (> \$5 billion) companies scored lower (0.15 to 0.20). Feasibility increased gradually with capitalization, peaking at 0.80 in the \$5 to \$10 billion range. Twelve feasibility ranges were defined to allocate resources according to companies' capacity to participate in the ecosystem. Sustainability evaluates ESG fit on a 0 to 1 scale, based on ESG ratings from agencies such as Sustainalytics (https://www.sustainalytics.com/). Sectors with lower ESG ratings were assigned lower sustainability scores, allowing the CEO to prioritize companies aligned with sustainable values.

These DSF scores inform the XGBoost model for synergy calculation. They are not part of the objective function but act as constraints that simulate strategic filters a CEO might impose during ecosystem configuration. This allows the model to explore trade-offs between financial

performance and strategic alignment. After completing the dataset, we split it into 100 projects for the Pyomo-Gurobi model and 50 projects for training the machine learning model. Although larger datasets could be used for optimization, in this first set of experiments we limit the selection to 100 projects to reflect a typical portfolio size in corporate ecosystem planning.

To validate the effectiveness of the proposed framework, we designed a series of computational experiments under both idealized and realistic conditions. Experiments 1 and 2, named 'static' and 'synergies' respectively, represent idealized scenarios without synergies (static) or with synergies estimates provided by the machine learning model. Experiments 3 and 4 extend the static and synergy scenarios by using the 'true' synergy values generated by a black-box function. In our experiments, this function is defined through a set of rules that compare pairs of assets along three synthetic attributes: desirability, sustainability, and feasibility. Each attribute takes values in the interval $[0, 1]$, and its mean over the training data is used as a reference point. Starting from a baseline synergy score of 1, the rules update the score as follows: (i) when both assets are above the mean, the synergy increases by 0.4; (ii) when both are below, the synergy decreases by 0.4; (iii) when one asset is above the mean and the other is below, the synergy increases by 0.3 if the first asset is above the mean and the second below, and decreases by 0.3 in the opposite case; (iv) when one asset is exactly equal to the mean and the other is above or below, the synergy is adjusted by ± 0.2, increasing if the other asset is above the mean and decreasing if it is below; and (v) when both assets are exactly equal to the mean, the adjustment depends on the magnitude of the mean itself, increasing by 0.2 if the mean is greater than 0.5 and decreasing by 0.2 otherwise. These updates are applied independently for each of the three attributes, and the resulting adjustments are aggregated to obtain the overall synergy score between the two assets.

6. Descriptive analysis of results

Fig. 1 shows the static Pareto frontier as a solid blue line, subject to the constraint of selecting between 3 and 7 projects. It indicates the trade-off between risk (standard deviation) and expected return in project portfolio selection without considering synergies. Each point on the curve represents an optimal portfolio for a given return level. Constructing this Pareto frontier required solving the optimization problem multiple times for different return thresholds to obtain a set of optimal solutions. We solved the problem for 400 distinct return values ranging from 0.07 to 0.15, minimizing risk at each level. This resulted in 400 optimal portfolios, each represented by a point on the frontier. Portfolios on the curve are efficient, while those below it are inefficient since higher returns can be achieved for the same risk. The curve clearly indicates the typical trade-off between risk and return.

The solid green line in Fig. 1 represents the synergies Pareto frontier. It corresponds to the portfolio optimized using estimated synergy effects, with inputs provided by the machine learning model prior to optimization. Notice that incorporating synergies shifts the Pareto frontier upward compared to the static case. In the static scenario, the synergy values are set to a neutral value of one. When project combinations are considered, positive or negative synergy effects adjust these values. For example, for the project pair p_4, p_8, the synergy value becomes $1 + 0.20$. A positive synergy is linked to an increase in the portfolio return, while a negative synergy decreases it, thus affecting the overall return. The difference between the two curves illustrates how incorporating predicted synergies enables more efficient portfolio configurations by capturing hidden interaction effects between projects. This approach allows CEOs to prioritize project combinations that generate collaborative benefits, while also aligning portfolio choices with strategic DSF goals.

Fig. 1 also shows, as a blue-dot line, the Pareto frontier for the static-adjusted scenario, where the neutral return values from the Gurobi solutions in experiment 1 are updated with the true returns generated by the black-box function, now accounting for real-life synergy effects. The

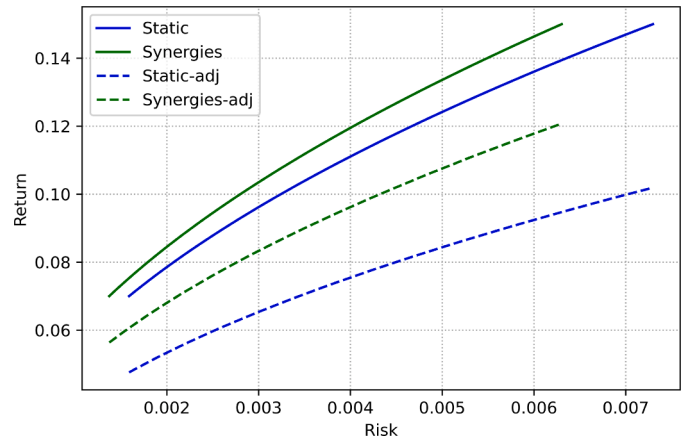

Fig. 1. Pareto frontiers for each experiment.

blue-dot curve illustrates the performance of the static portfolio when true synergies are employed. As observed, the static-adjusted portfolio yields lower returns for a given level of risk compared to the original static curve. This downward shift suggests that portfolios constructed without considering synergies may easily be suboptimal. This makes sense, as ignoring synergies in project portfolio optimization leads to suboptimal selections, since the model overlooks how projects interact to affect returns. Conversely, when synergy effects can be effectively estimated, the optimization component can identify project combinations that enhance positive interactions and avoid negative ones, yielding portfolios with higher returns for a given level of risk.

Finally, the green-dot line shows the Pareto frontier for the synergies-adjusted scenario, where the solutions provided by Gurobi in experiment 2 are updated with the true synergies instead of the estimated ones. As expected, the synergies-adjusted portfolio still offers a more favorable risk-return profile than the static-adjusted portfolio (blue-dot line), since the machine learning model provides more accurate inputs to the optimization component. As the estimated synergies approach their true values, the green-dot line converges toward the original green line from experiment 2. This highlights the importance of the machine learning model's accuracy for obtaining near-optimal project portfolio configurations and justifies the choice of a highly efficient regressor, such as XGBoost, over less effective alternatives like linear regressors that might not capture complex patterns in the data.

7. Statistical analysis

To evaluate whether the return differences across the four portfolio strategies are statistically significant, we performed an analysis of variance (ANOVA), followed by pairwise comparisons using Fisher's Least Significant Difference (LSD) test with Bonferroni correction. Interpolated returns were computed at nine target risk levels for each of the following strategies: static, synergies, static-adjusted, and synergies-adjusted. These interpolated values were organized with the risk level treated as a blocking variable to account for repeated measures. The ANOVA results (Table 2) indicate a significant main effect of the portfolio strategy on returns, suggesting that at least some strategies yield statistically different results. The effect of risk level was also significant, as expected.

Table 2
ANOVA results for repeated measures on return across scenarios.

Source	Sum Sq	df	F	p-value
C(Scenario)	0.009787	3	282.114	0.0000
C(Risk)	0.010022	8	108.342	0.0000
Residual	0.000278	24	–	–

Table 3
Fisher LSD pairwise comparisons (Bonferroni corrected).

Group 1	Group 2	t-stat	p-value (corrected)	Significant
Static	Static-adjusted	4.4572	0.0024	Yes
Static	Synergies	−0.8688	1.0000	No
Static	Synergies-adjusted	1.7015	0.6492	No
Static-adjusted	Synergies	−5.2379	0.0005	Yes
Static-adjusted	Synergies-adjusted	−2.8511	0.0693	No
Synergies	Synergies-adjusted	2.5538	0.1274	No

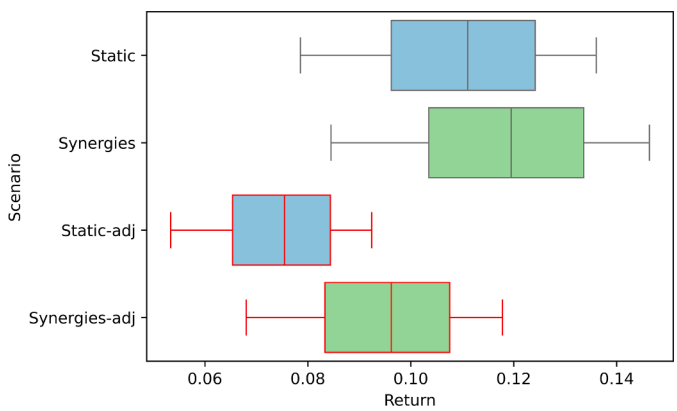

Fig. 2. Return distribution by portfolio strategy.

We then applied Fisher's LSD test to compare each pair of strategies. Bonferroni correction was applied to account for multiple testing. The results are shown in Table 3. Only two comparisons yielded statistically significant differences after correction: (i) static vs. static-adjusted returns differ significantly (corrected $p = .0024$); and (ii) static-adjusted vs. synergies returns differ significantly (corrected $p = .0005$). No significant differences were found between static and synergies, nor between synergies and synergies adjusted.

These findings show that adjusting the static portfolio for risk leads to significantly lower returns compared to both the unadjusted static and the synergy-based strategies. These results align with the earlier observation that the static approach, which does not consider synergies, might yield suboptimal solutions. Fig. 2 presents the distribution of returns across strategies. The static-adjusted strategy shows a notably lower return distribution compared to the others, which also aligns with the ANOVA results.

8. Scalability and sensitivity analyses

In addition to the proof-of-concept experiments conducted with 100 projects, we assessed the scalability of the proposed hybrid methodology by testing its performance on larger project portfolio optimization problems, considering both synergy effects and strategic factors. The aim of this analysis was to understand how increasing the number of projects affects computational time and memory usage, and to determine whether the optimization solver (Gurobi) could handle larger problem instances without exhausting system resources. To perform these experiments, we increased the number of projects incrementally from 100 up to 500. Specifically, we considered portfolio sizes of 100, 200, 300, 400, and 500 projects. To accommodate the increased computational demand, all experiments were executed on a high-performance machine with the following specifications: Linux Ubuntu 24.04.2, AMD Ryzen Threadripper 7970X CPU running at 5.356 GHz, and 128 GB of RAM. The results obtained are summarized in Fig. 3. Both memory usage (RAM) and computational time increase with the size of the portfolio, as expected. For 100 projects, the solver required approximately 192 MB of RAM and 3.5 seconds of computation time. Increasing the number of projects to 500 led to a memory consumption of around 1.32 GB and a

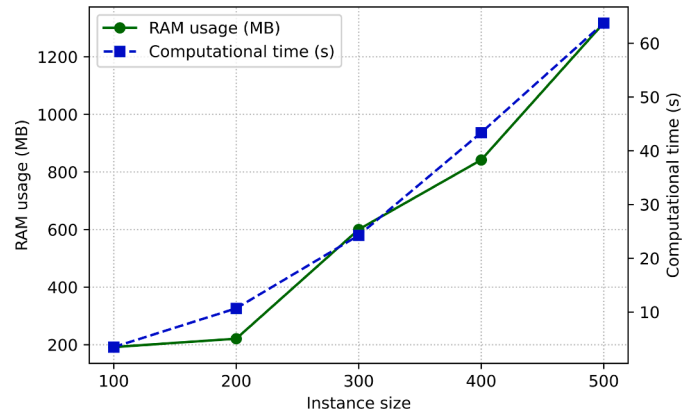

Fig. 3. Scalability analysis of the proposed methodology.

computation time of approximately 64 seconds. Due to this moderate RAM usage, Gurobi successfully solved all instances without exhausting memory.

The observed trend indicates that both memory requirements and computation times increase with the number of projects in a nonlinear manner. Specifically, the growth appears to be faster than linear but slower than cubic. Even for 500 projects, the absolute levels of RAM (around 1.3 GB) and computation time (approximately 64 seconds) remain moderate, well within the capabilities of a modern computer. While smaller instances (up to 200 projects) are solved very quickly and with minimal RAM usage, larger instances (above 400 projects) require more computational resources. Nevertheless, the methodology remains tractable for portfolios of up to 500 projects. This analysis confirms that the proposed hybrid methodology can be effectively applied beyond the initial proof-of-concept scale, enabling managers to consider larger ecosystems with complex interactions and strategic constraints while maintaining feasible computational performance.

To evaluate the robustness of the proposed hybrid methodology, we conducted a sensitivity analysis focusing on the impact of the strategic desirability, feasibility, and sustainability factors. Specifically, we assessed how small variations in the DSF values affect the return-risk Pareto frontier under the synergy scenario. The original frontier was first computed using the baseline DSF constraints. We then repeated the optimization process by relaxing the DSF values by 5 %, effectively allowing less restrictive strategic constraints. This produced an upper bound frontier, as portfolios could include projects with slightly weaker DSF scores while still meeting the relaxed thresholds. Finally, we applied the opposite perturbation by tightening the DSF values by 5 %, thus requiring stricter strategic alignment. This produced a lower bound frontier, reflecting more demanding conditions. The resulting three curves are shown in Fig. 4. As expected, the baseline synergy curve lies between the two perturbed frontiers. The upper bound frontier dominates the baseline, yielding higher returns for comparable levels of risk because the relaxation allows more projects to qualify despite weaker DSF attributes. Conversely, the lower bound frontier shifts downward, with lower achievable returns at the same risk levels due to the exclusion of otherwise feasible projects that fail to meet the stricter DSF requirements.

Notice that the gap between the upper and lower bound curves is relatively modest across the entire return-risk spectrum, suggesting that the methodology is not overly sensitive to small perturbations in DSF values. Furthermore, even under stricter DSF conditions the optimization process still identifies portfolios with positive synergies and competitive returns, confirming the robustness of the proposed methodology. This robustness is essential for practical decision-making, as DSF values in real-world settings are often estimated with some degree of uncertainty.

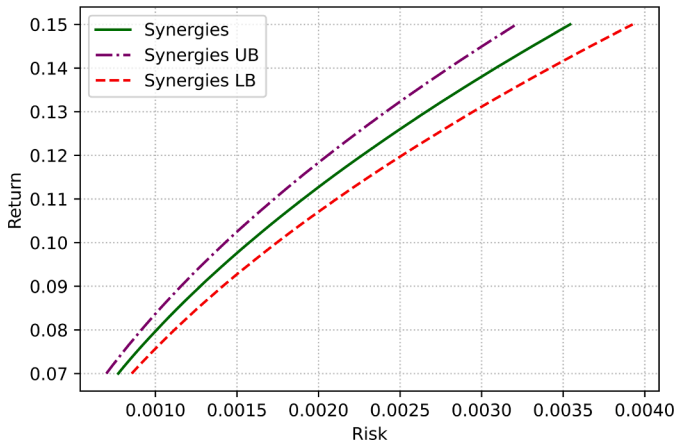

Fig. 4. Sensitivity analysis of the proposed methodology.

9. Managerial insights

Based on the previous discussion, adopting synergy-aware optimization tools offers new opportunities for CEOs and corporate ecosystem managers. The proposed hybrid methodology enables more informed strategic decision-making. The machine learning component predicts synergy coefficients, improving the quality of input for optimization (Karimi-Mamaghan et al., 2022). Analyzing potential synergies allows AI to identify combinations of business units or partnerships that can yield higher returns, thus promoting innovation and new market opportunities (Clarysse et al., 2014). This supports more efficient resource allocation and sharper focus on projects that promote sustainable growth. To implement such a methodology, organizations must ensure access to high-quality, structured data on project characteristics, interdependencies, and ecosystem context. This includes data on objectives, resource requirements, strategic alignment, and historical performance. In practice, this implies investing in a reliable data infrastructure that supports real-time data collection, processing, and sharing across ecosystem participants.

Another benefit of AI-based decision tools is the mitigation of cognitive biases. Algorithmic support provides objective, data-driven insights that reduce the risk of misjudging project value. This increases the reliability of strategic decisions and helps build more effective ecosystems. However, this also requires active engagement by CEOs and senior managers, including regular communication with data teams and a willingness to incorporate analytical tools into strategic planning. Beyond technical integration, using AI for ecosystem design involves a cultural shift. Executives may need to adjust decision-making practices to include model-based recommendations, especially when dealing with trade-offs. Building trust in the model's outputs is essential. This can be supported by explanations of how synergy scores are calculated and how they influence portfolio optimization. Because synergies evolve over time, ecosystem strategies must remain adaptive. Changes in market conditions, technologies, or partnerships may alter the strength and nature of synergies. This requires periodic updates to the model and regular re-evaluation of the portfolio configuration.

Despite the potential, there are several challenges to consider. First, modeling synergies is inherently complex. Synergies can result from multiple sources, i.e.: technological compatibility, market overlap, or operational integration. Poor synergy modeling may lead to suboptimal or misaligned portfolio choices. Second, the success of the model depends on data quality and availability. Incomplete or noisy data can undermine the effectiveness of the methodology. Lastly, synergies are dynamic and subject to change coinciding with market trends, regulatory shifts, and evolving strategic goals.

10. Conclusion and future work

This paper proposes a hybrid methodology to help CEOs design corporate ecosystems by combining AI-based synergy prediction with project portfolio optimization. Our approach integrates exact optimization techniques with machine learning to support decision-making in complex environments. Including AI-predicted synergies alters the composition of optimal portfolios compared to static models, highlighting the importance of modeling project interactions. Considering synergy effects, rather than ignoring them, allows the optimization module to identify combinations that benefit from positive synergies and yield improved returns. In addition, incorporating strategic criteria (desirability, feasibility, and sustainability) helps align financial and strategic goals. The proposed methodology has practical relevance for managers involved in development, strategic planning, innovation, or investment. It provides a structured way to account for strategic DSF factors and synergy effects, which are often overlooked in traditional portfolio models.

In our computational experiments, we used a combination of real-life and synthetically generated data to maintain clarity and ease of use, but the methodology is sufficiently general to apply to any corporate dataset with historical observations on DSF levels and synergy effects. The proof-of-concept experiments were implemented in Python using Pyomo, Gurobi, and XGBoost, and applied to data from Nasdaq. Strategic filters were introduced through DSF scores to simulate realistic CEO constraints. The experimental analysis compared four scenarios: static (no synergies), synergies (machine learning-predicted), static-adjusted (true synergies applied post-optimization), and synergies-adjusted (true synergies replacing predicted ones). Pareto frontiers showed that incorporating synergies shifted the efficient frontier upward, improving returns for the same level of risk. In contrast, static portfolios adjusted with true synergies underperformed, confirming that ignoring interdependencies can lead to suboptimal allocations. The synergies-adjusted portfolios retained a more favorable risk-return profile, showing that accurate synergy estimation can meaningfully improve portfolio efficiency. An ANOVA test confirmed these findings. Scalability tests further showed that the methodology can easily handle up to 500 projects, with memory use of just about 1.3 GB and computation time of 64 seconds, confirming feasibility for larger applications. Sensitivity analysis also revealed that relaxing or tightening DSF constraints by 5 % modestly shifted the Pareto frontier but did not alter the overall conclusions, indicating certain robustness under parameter uncertainty.

While the results are promising, several limitations arise. First, this is a conceptual experiment rather than a full deployment solution. Real-world application would require guidance on data collection, parameter tuning, system integration, real-time updates, and risk management. Second, although the optimization algorithm finds optimal solutions for given parameters, the synergy predictions from the XGBoost model are uncertain and may not fully reflect real-world variability. The results should therefore be viewed as indicative. Third, the model was validated using data from Nasdaq-listed firms, which ensures good disclosure quality but may limit generalizability to less uniform settings, such as emerging markets or private firms. Finally, the DSF parameters used to train the model were synthetically generated. Although designed to reflect CEO decision logic, they are not empirically derived. Future research should focus on capturing the temporal evolution of synergies using time-series analysis or reinforcement learning, and on validating the methodology in real corporate ecosystems.

CRediT authorship contribution statement

Patricia Rodriguez-Garcia: Conceptualization,Methodology, Formal analysis, Investigation, Writing – original draft; **Angel A. Juan:** Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – review & editing, Supervision; **Jon A. Martin:** Software; **David Lopez-Lopez:** Conceptualization, Validation, Writing – review & editing, Supervision; **Josep M. Marco:** Supervision.

Data availability

Link to github.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.eswa.2025.129593.

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